

Hall Effect Principle Diagram

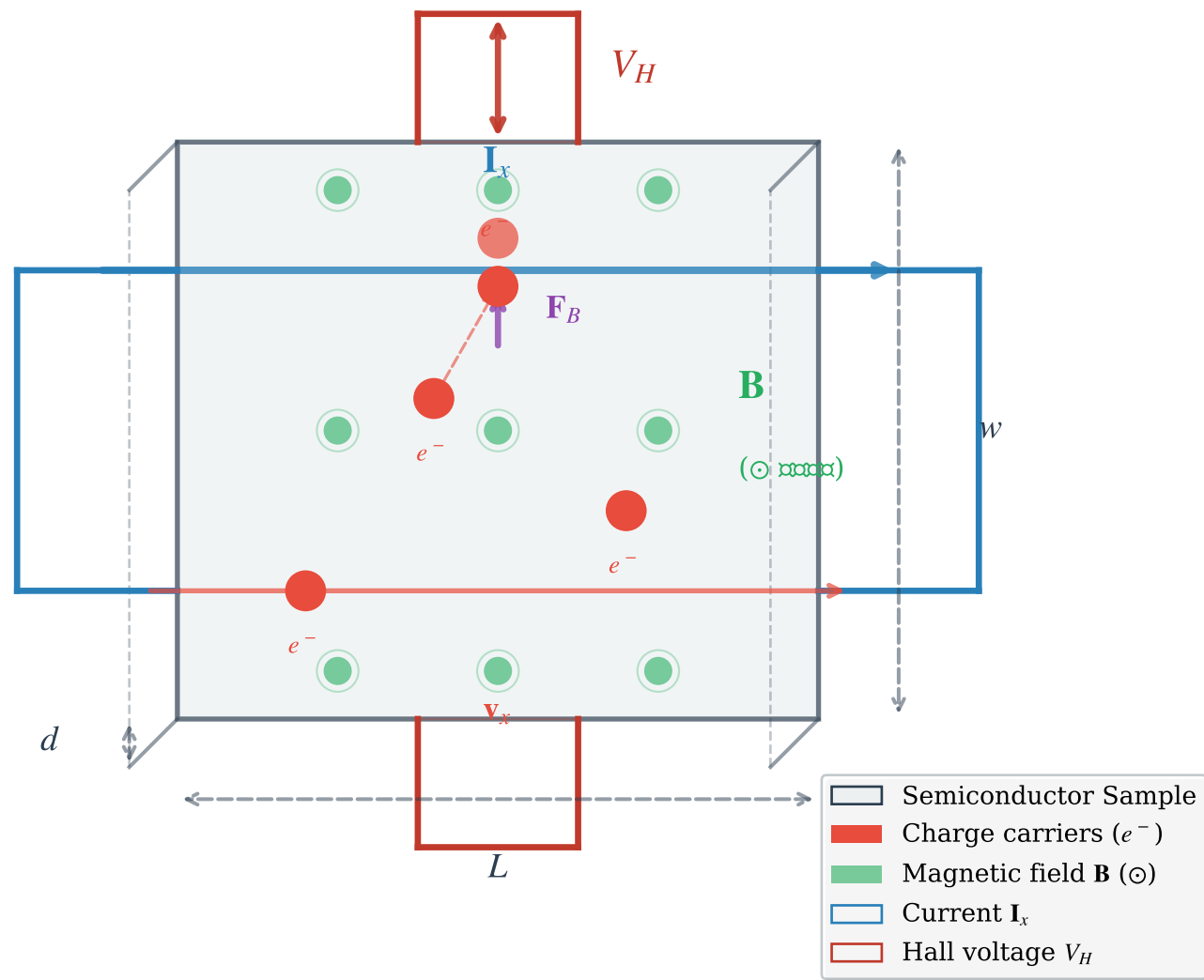


Fig. 1 Schematic diagram of the Hall effect with corresponding mathematical derivation.

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Mathematical Derivation

(1) Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

(2) Steady state ($\mathbf{F} = 0$):

$$\mathbf{E}_H = -\mathbf{v}_x \times \mathbf{B}_z$$

(3) Hall electric field:

$$E_H = v_x B_z$$

(4) Hall voltage:

$$V_H = E_H \cdot w = v_x B_z w$$

(5) Drift velocity:

$$v_x = \frac{I_x}{nqwd}$$

• Hall effect: $V_H \propto B_z$

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(6) Hall coefficient:

$$R_H = \frac{E_H}{J_x B_z} = \frac{1}{nq}$$

(7) Final expression:

$$V_H = R_H \frac{I_x B_z}{d} = \frac{I_x B_z}{nqd}$$

where:

n = carrier density, q = elementary charge

d = sample thickness, w = sample width

J_x = current density, R_H = Hall coefficient